

input $x \in \mathbb{R}$ (one number)

input $\mathbf{x} \in \mathbb{R}^n$ (n numbers)

output $f \in \mathbb{R}$
(one number)



Case 1: number (1×1)
 $f'(3) = 6$

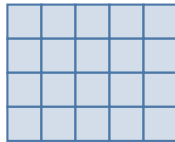


Case 2: row ($1 \times n$)
gradient $\nabla f(2, 3) = [4, 6]$

output $\mathbf{f} \in \mathbb{R}^m$
(m numbers)



Case 3: column ($m \times 1$)
 $\mathbf{g}'(1) = [1, 2]^\top$



Case 4: matrix ($m \times n$)
Jacobian J